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Modelling with the derivative

Four lessons of real situations — motion, growth, cost, flow — in which the derivative is the quantity the question is really about.

LESSON 31–34 | 4 × 40 MIN | ALGEBRA 11, §2.2 | P1 · 9.4

LESSON CLOCK

16 | 30' | 30' | 30' | 50' | 24'

The derivative has units	16 min
Motion	30 min
Connected rates	30 min
Cost and revenue	30 min
Practice	50 min
Homework and consolidation	24 min

LEARNING OBJECTIVES

- Interpret a derivative as a rate of change in a stated context, with units.
- Model motion with displacement, velocity and acceleration.
- Use connected rates of change.
- Interpret marginal cost and marginal revenue.

TEXTBOOK

National	pp. 147–168
Cambridge	Pure Mathematics 1, pp. 186–190
Workbook	P1 Exercise 9C

01 Explanation

The derivative has units

READ THE UNITS OFF THE FRACTION

$\frac{dy}{dx}$ carries the units of y divided by those of x. A distance in metres over a time in seconds gives m/s; a cost in so‘m over a number of items gives so‘m per item. Stating the units is half the interpretation.

Y	X	$\frac{DY}{DX}$	UNITS
distance	time	velocity	m/s
velocity	time	acceleration	m/s^2
volume	time	flow rate	cm^3/s
total cost	quantity	marginal cost	soʻm per item
population	time	growth rate	people per year

Motion

$$v(t) = s'(t) \quad a(t) = v'(t) = s''(t)$$

QUESTION IN WORDS	WHAT TO SOLVE
when is the body at rest?	$v(t) = 0$
when does it change direction?	v changes sign
when is the speed greatest?	$a(t) = 0$
how far has it travelled?	total distance — count each direction separately

DISPLACEMENT IS NOT DISTANCE

A body that goes out 5 m and comes back has displacement 0 and distance 10 m. If v changes sign inside the interval, the journey must be split at that instant.

Connected rates

When two quantities are linked by a formula, their rates are linked by the chain rule:

$$\frac{dA}{dt} = \frac{dA}{dr} \cdot \frac{dr}{dt}$$

Example. Water is poured into a cone at 20 cm³/s. The cone has height 30 and top radius 15 cm. How fast is the depth rising when $h = 10$?

$$r = \frac{h}{2} \Rightarrow V = \frac{1}{3}\pi\left(\frac{h}{2}\right)^2 h = \frac{\pi h^3}{12}$$

$$\frac{dV}{dh} = \frac{\pi h^2}{4} = 25\pi \Rightarrow \frac{dh}{dt} = \frac{20}{25\pi} \approx 0.255 \text{ cm/s}$$

ALWAYS REDUCE TO ONE VARIABLE FIRST

The relation $r = \frac{h}{2}$ comes from similar triangles and must be used **before** differentiating. Differentiating a two-variable formula is the standard error here.

Cost and revenue

In economics the derivative of a total is called a **marginal** quantity: the approximate cost of producing **one more** item.

$$MC(q) = C'(q) \quad MR(q) = R'(q) \quad \text{profit is greatest when } MC = MR$$

The last statement is the optimisation of Lesson 23–25 in economic dress: profit $P = R - C$ is stationary when $P' = R' - C' = 0$.

Example. $C(q) = 0.02q^2 + 8q + 500$ so‘m. The marginal cost at $q = 100$ is $C'(100) = 0.04 \times 100 + 8 = 12$ so‘m — the 101st item costs about 12 so‘m to make, while the average cost of the first hundred is $\frac{1500}{100} = 15$ so‘m.

02 Worked examples

EXAMPLE 1 **$s = t^3 - 6t^2 + 9t$ metres. Find when the body is at rest and the total distance in the first 4 s.**

1. $v = 3t^2 - 12t + 9 = 3(t - 1)(t - 3)$

   Rest at $t = 1, 3$.

2. $s(0) = 0,\ s(1) = 4,\ s(3) = 0,\ s(4) = 4$

3. Distances: $4 + 4 + 4$.

   Direction changes twice.

ANSWER

Rest at $t = 1$ and $t = 3$; distance 12 m

EXAMPLE 2 **A ladder 5 m long slides down a wall. Its foot moves out at 0.4 m/s. How fast is the top falling when the foot is 3 m out?**

1. $x^2 + y^2 = 25$

   Differentiate with respect to t.

2. $2x\dfrac{dx}{dt} + 2y\dfrac{dy}{dt} = 0$

3. $x = 3 \Rightarrow y = 4$

4. $3(0.4) + 4\dfrac{dy}{dt} = 0 \Rightarrow \dfrac{dy}{dt} = -0.3$

ANSWER

0.3 m/s downwards

EXAMPLE 3 **$C(q) = 0.05q^2 + 12q + 800$. Find the marginal cost at $q = 60$ and the average cost.**

1. $C'(q) = 0.1q + 12$
2. $C'(60) = 18$ so‘m per item.
3. $C(60) = 180 + 720 + 800 = 1700$
4. Average $\frac{1700}{60} \approx 28.3$ so‘m.

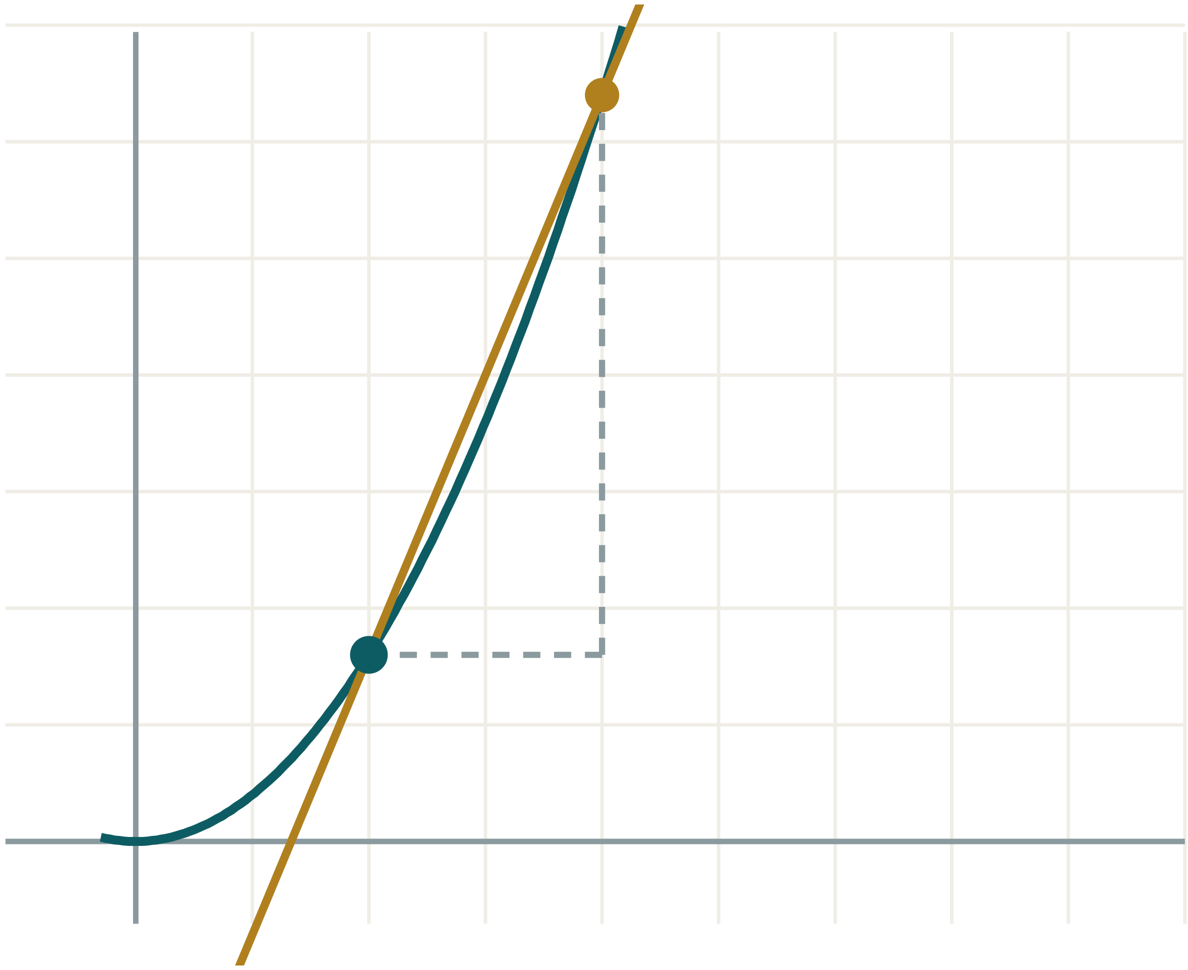
ANSWER

Marginal 18, average ≈ 28.3 so‘m

03 Interactive model

Give every answer with its units, out loud, before writing it down.

A rate of change



x **2** | h **2** | Δy / Δx **2.4** | f ′(x) exactly **1.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Mathematical model	Matematik model	Математическая модель
Rate of change	O‘zgarish tezligi	Скорость изменения
Displacement	Ko‘chish	Перемещение
Velocity	Tezlik	Скорость
Acceleration	Tezlanish	Ускорение
Connected rates	Bog‘liq tezliklar	Связанные скорости
Marginal cost	Marjinal xarajat	Предельные издержки
Marginal revenue	Marjinal daromad	Предельный доход
Units of a rate	Tezlik birligi	Единицы скорости
Instant of rest	To‘xtash payti	Момент остановки

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

If s is in metres and t in seconds, s' is in:

- m
- m/s
- m/s^2
- s/m

A body is at rest when:

- $s = 0$
- $v = 0$
- $a = 0$
- $s'' = 0$

Marginal cost is:

- $C(q)$
- $C'(q)$
- $\frac{C(q)}{q}$
- $C''(q)$

Connected rates use:

- the product rule
- the chain rule
- the quotient rule
- first principles

Profit is greatest when:

- $MC = 0$
- $MR = 0$
- $MC = MR$
- $C = R$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $s = 5t^2$; find $v(3)$

   ANSWER 30 m/s

2. $s = t^3$; find $a(2)$

   ANSWER 12 m/s²

3. $s = 4t - t^2$; when is the body at rest?

   ANSWER $t = 2$

4. Units of $\frac{dV}{dt}$ for volume in cm³

   ANSWER cm^3/s

5. $C(q) = 3q + 50$; marginal cost

   ANSWER 3

6. $V = \frac{4}{3}\pi r^3$; find $\frac{dV}{dr}$

   ANSWER $4\pi r^2$

7. $A = \pi r^2$; find $\frac{dA}{dr}$ at $r = 3$

   ANSWER 6π

Medium · the standard question

7 PROBLEMS

1. $s = t^3 - 6t^2 + 9t$; when at rest?

   ANSWER $t = 1, 3$

2. Same; total distance in the first 4 s

   ANSWER 12 m

3. Balloon: $\frac{dr}{dt} = 2$ cm/s; find $\frac{dV}{dt}$ at $r = 5$

   ANSWER 200π cm³/s

4. Circle: $\frac{dr}{dt} = 0.5$; find $\frac{dA}{dt}$ at $r = 8$

   ANSWER 8π

5. Ladder 5 m, foot at 0.4 m/s; top's speed when foot is 3 m out

   ANSWER 0.3 m/s

6. $C(q) = 0.05q^2 + 12q + 800$; marginal cost at $q = 60$

   ANSWER 18

7. $s = 20t - 5t^2$; greatest height and when

   ANSWER 20 m at $t = 2$

Hard · stretch and reason

7 PROBLEMS

1. Cone, height 30, radius 15, filled at 20 cm³/s. $\dfrac{dh}{dt}$ at $h = 10$

   ANSWER ≈ 0.255 cm/s

2. Same cone at $h = 20$

   ANSWER ≈ 0.064 cm/s

3. $s = t^4 - 8t^2$; find all instants of rest

   ANSWER $t = 0, \pm 2$

4. A cube's volume grows at 12 cm³/s. Find $\dfrac{da}{dt}$ when $a = 4$

   ANSWER 0.25 cm/s

5. Same cube: find $\dfrac{dS}{dt}$ at that moment

   ANSWER 12 cm²/s

6. $R(q) = 60q - 0.5q^2$, $C(q) = 10q + 200$. Maximise the profit

   ANSWER $q = 50$, profit 1050

7. A trough 10 m long, cross-section an isosceles triangle 2 m wide and 1 m deep, fills at 0.5 m³/min. Find $\dfrac{dh}{dt}$ at $h = 0.5$

   ANSWER 0.05 m/min

07 Homework

Homework — 6 tasks

Every answer needs its units and a sentence in the words of the question.

1. $s = 2t^3 - 15t^2 + 24t$. Find when the body is at rest, and the total distance in the first 5 s.
2. A spherical balloon is inflated at 30 cm³/s. Find how fast the radius grows when $r = 5$ cm.

3. A ladder 13 m long slides; the foot moves at 0.5 m/s. Find the top's speed when the foot is 5 m out.
4. $C(q) = 0.1q^2 + 20q + 1000$. Find the marginal cost at $q = 40$ and the average cost.
5. $R(q) = 100q - q^2$ and $C(q) = 20q + 300$. Find the profit-maximising q.
6. A conical tank of height 12 m and top radius 6 m is filled at 3 m³/min. Find $\frac{\mathrm{dh}}{\mathrm{dt}}$ when $h = 4$ m.